

Steven Gerard Mascarenhas

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SUMMARY

Software engineer with an MSc in AI and ML and experience building **LLM workflows**, **API integrations**, and cloud-deployed models on AWS. Built an **AI medical RAG chatbot** (Python, FastAPI, SQL, Docker, CI/CD), achieved **90%+ accuracy** on a skin-lesion classifier with SageMaker, and shipped full-stack apps with React and REST APIs. Won **Best Solana Solution** at Ireland's first Blockchain × Quantum hackathon. Strong **ownership mindset** with a collaborative, build-fast approach.

EXPERIENCE

Software Quality Assurance Associate | Trigyn Technologies, Mumbai, India Dec 2023 – Aug 2024

- Reduced project delivery delays on the UN Mental Health project through project management and process compliance audits, aligned with ISO software quality standards, across Agile and Waterfall workflows.
- Designed and executed **200+ test cases** across functional and regression testing, logging **100+ defects** in JIRA and monitoring code quality with SonarQube before release.
- Decreased rework and defect rates across a team of **20+ engineers** by performing Root Cause Analysis (RCA) on recurring defects and leading knowledge-sharing sessions on risk management.

Full Stack Software Engineer Intern | Patnekar Private Limited (PPL), Goa, India Oct 2022 – Jul 2023

Built a hyperlocal marketplace connecting local businesses to customers across 100+ categories.

- Built the customer side serving **2,000+ users**, with **10+ responsive web pages** and REST APIs using CodeIgniter MVC, PHP, JavaScript, JQuery, SCSS, and MySQL on AWS RDS.
- Implemented JWT authentication for the customer REST API, with ZeroBounce API for email validation at signup.
- Built the cart, checkout, and order tracking flow integrating Cashfree for payments, Google Maps API for location-based vendor discovery and tiered delivery options, and Amazon SES for order confirmation emails.
- Worked in a **4-developer team** using Agile Scrum across **6+ sprints**, with code reviews, peer testing, and Trello for backlog tracking and sprint management.
- Maintained **99.9%** platform uptime and reliable production deployments by setting up AWS infrastructure (EC2, RDS, S3, IAM) and CloudWatch monitoring, and automated CI/CD across multi-stage environments using AWS CodePipeline.

Software Developer Intern | Pai Kane Group, Goa, India Apr 2021 – Jun 2022

Built the admin dashboard for ServeME, a platform to digitize generator servicing and inventory.

- Saved **10+ hours per week** and digitised manual workflows for **1,000+ genset products** by building a real-time inventory and service-management dashboard with **8 CRUD modules** on React, CoreUI, and Firebase.
- Built a scheduling module linking each service request to its customer, genset, assigned technician, and required spare parts with real-time status tracking, secured with Firebase Authentication.
- Identified and fixed recurring bugs across the React frontend and Firebase backend through systematic debugging.

Machine Learning Engineer Intern | Cloud Counselage Pvt Ltd, Mumbai, India Jul 2021 – Sept 2021

- Achieved an **RMSE of 1.05**, validated by 5-fold cross-validation, by building an item-based collaborative filtering movie recommender in Python using KNN with cosine similarity on the MovieLens 100K dataset.
- Predicted Amazon stock prices over a **14-day forecast** with an **RMSE of \$3.08** by training an LSTM model in PyTorch.

PROJECTS

Link: Hackathon Winner: Prism - Crypto-Agile Post-Quantum Vault for Solana | Superteam Ireland May 2026

Won Best Solana Solution and 3rd Place Overall (70+ participants) at the world's first Blockchain × Quantum Hackathon.

- Built **Prism**, a crypto-agile migration layer that protects Solana assets from quantum attacks through a smart vault integrated with a decentralised post-quantum validation system, supporting NIST-standardised Falcon and ML-DSA (FIPS-204) algorithms.
- Tech: Rust, Anchor, Solana Devnet, Falcon-512, ML-DSA-44, Open Quantum Safe, Phantom Wallet, TypeScript, React.*

GitHub Link: End-to-End AI Medical Chatbot - RAG Pipeline with Cloud Deployment

Mar 2026

- Built a production RAG pipeline over a **637-page** medical corpus (**5,800+ chunks**, HuggingFace embeddings, Pinecone vector store) with reranking, serving real-time responses with FastAPI and React across Groq LLaMA 3.1 and OpenAI GPT-4o, containerised with Docker/ECR and deployed to AWS EC2 with Terraform and GitHub CI/CD.
- Evaluated and monitored the pipeline with LangSmith, scoring **100%** on four LLM-as-judge metrics, with request tracing for latency, token usage, and cost.
- *Tech used: Python, LangChain, Pinecone, Cohere Rerank, FastAPI, HuggingFace, Groq, OpenAI, LangSmith, Docker, Terraform, AWS EC2, ECR, GitHub Actions CI/CD.*

GitHub Link: Skin Lesion Classification for Skin Cancer Detection | University of Limerick

May 2025

- Achieved **90%+ accuracy** classifying **7 lesion types** across **10,000+ images** using a pretrained HuggingFace Vision Transformer and automated the pipeline (S3, Lambda, SageMaker, CloudWatch, IAM-secured access) with a Streamlit front-end for instant diagnoses.
- *Tech used: AWS SageMaker, HuggingFace, AWS Lambda, S3, IAM, CloudWatch, Streamlit.*

Project Link: Hackathon: Smart Pay - AI BNPL Recommender for SME merchants | Hack Ireland

Feb 2025

- Built a browser extension using the OpenAI API and prompt engineering to rank 3 providers for personalised recommendations.
- *Tech used: OpenAI API, FastAPI, PostgreSQL, React.*

TECHNICAL SKILLS

- **AI & Machine Learning:** LLMs, RAG Pipelines, Vector Database (Pinecone), LangChain, Prompt Engineering, Image Processing, Computer Vision, Collaborative Filtering, Hugging Face Transformers, PyTorch, TensorFlow, Scikit-learn, XGBoost, LSTM, Deep Learning, NLP, Pandas, NumPy, Matplotlib, OpenAI, Groq, Claude.
- **Cloud & DevOps:** AWS (EC2, ECR, S3, Lambda, SageMaker, CloudWatch, IAM, CodePipeline), Docker, Terraform (IaC), Git, GitHub Actions CI/CD.
- **Languages & Frameworks:** Python, JavaScript, TypeScript, C++, PHP, SQL (PostgreSQL, MySQL), HTML, CSS/SCSS, React, React Native, Node.js, FastAPI, JQuery, CodeIgniter MVC, Bootstrap, REST APIs.
- **Engineering Practices:** Unit Testing, Integration Testing, Functional & Regression Testing, Test Case Design, Defect Tracking (JIRA), SonarQube, Code Reviews, Debugging, Root Cause Analysis, Agile/Scrum, SDLC.

EDUCATION

MSc in Artificial Intelligence and ML | University of Limerick

Sept 2024 – Aug 2025

QCA: 3.31/4, Second Class Honours (2.1), Grade 1

Bachelor in Computer Engineering | Goa University

Jul 2019 – Jun 2023

CGPA: 9.1/10, Grade: A+

PUBLICATIONS

Hyperlocal Marketplace: Transforming Unorganized Localized Marketplace with Future Tech

Dec 2023

S. G. Mascarenhas et al., IEEE 5th International Conference on Advances in Computing, Communication Control and Networking (ICAC3N), 2023.

- Co-authored a peer-reviewed IEEE paper proposing a tech-enabled platform to digitize unorganized local marketplaces and connect local businesses to customers.

CERTIFICATIONS

AWS Academy - Machine Learning Foundations

Apr 2025

AWS Academy - Cloud Foundations

Mar 2025